

Fabrication of liver microtissue with liver decellularized extracellular matrix (dECM) bioink by digital light processing (DLP) bioprinting

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ABSTRACT

As one of the most effective treatments of end-stage liver disease, liver transplantation still suffers from a shortage of donor organs or a low degree of engraftment. Thus, alternatives to liver transplantation, such as liver support systems, have to be extensively explored. In this study, a novel liver microtissue with an inner gear-like structure, which achieved a larger body surface area, was designed and manufactured to improve hepatic functional restoration. The liver-specific bioinks were developed by combining photocurable methacrylated gelatin (GelMA) with liver decellularized extracellular matrix (dECM), and human-induced hepatocytes (hiHep cells) were encapsulated to form cell-laden bioinks. The mechanical properties, swelling, and cytocompatibility of GelMA/dECM bioinks were carefully characterized before 3D printing. Then, the digital light process (DLP)-based bioprinting was used to fabricate the liver microtissue, and liver dECM was found to improve both the printability and cell viability of GelMA bioinks. hiHep cells were also found to spread farther and have better hepatocyte-specific functions (albumin secretion and urea) in the liver microtissue when liver dECM was added to the GelMA bioinks. Our results provide a promising liver dECM-based cell-laden bioink for liver microtissue fabrication, which would be a potential liver tissue engineering product to help restore hepatic functions.

1. Introduction

The liver is a vital organ because of its critical functions in protein synthesis and xenobiotic metabolism [1,2], but deaths from liver disease are increasing worldwide [3]. Although end-stage liver disease (ESLD) has been reported to be treated by liver or hepatocyte transplantation [4], these treatments are still limited by the shortage of donor organs or a low degree of engraftment [5]. Therefore, there is still a great challenge and uncertainty about the treatment of ESLD. Recently, bioengineering strategies, especially tissue engineering-based approaches, have emerged that are focused on developing transplantable liver microtissues or an entire bioengineered liver as alternatives [6] or long-term hepatic functional restoration [2]. A critical

prerequisite of successful tissue engineering products is the creation of a three-dimensional (3D) cell environment to overcome the limitations of two-dimensional (2D) cell culture [7]. Therefore, to achieve functional liver recovery, complex 3D structures similar to native tissues must be reconstructed during liver tissue engineering. Recently, the emerging technology of additive manufacturing has gained increasing attention in the fabrication of patient-specific scaffolds of tissue engineering [2]. 3D bioprinting provides the most promising advancement in precisely depositing various biomaterials and multiple cell types to manufacture biological structures in vitro.

Currently, several investigations have been reported using 3D printing in liver regenerative medicine [5]. For example, Lee et al. [6] used extrusion-based printing in liver tissue engineering and focused on

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developing a thermal cross-linkable liver decellularized extracellular matrix (dECM) bioink, where the functionality of human hepatocellular carcinoma (HepG2) cells (human hepatoma cell lines) in the liver dECM bioink was evaluated. Hiller et al. [8] also applied an extrusion-based printing approach to print human bipotent hepatic progenitor (HepaRG) liver cells with good viability and metabolic functions, where the bioink was obtained by adding human dECM into an alginate/gelatin-based bioink. A few trends are noticeable in these studies. First, the extrusion-based printing technique is often used, because extrusion-based printing is the most commonly explored approach among all the 3D bioprinting techniques due to its bioink versatility and operational convenience [9]. Second, liver dECM-based materials have been considered as the most biomimetic organ-specific bioinks [10], because a native ECM provides mechanical, biophysical, and biochemical cues to cells, and dECM-based bioinks can inherit the intrinsic cues from a native ECM [11]. Third, primary hepatocytes and HepG2 have been widely used in previous studies of liver tissue engineering, due to the fact that their functions are similar to native liver tissues [12,13].

Although achievements have been made using extrusion-based printing in liver tissue engineering, these studies still suffer from difficulties in low printing resolution and poor structural integrity [10] due to the inherent weakness of extrusion-based printability and insufficient mechanical properties of dECM-based bioinks [11]. To overcome these shortcomings in extrusion-based printing of dECM-based bioinks, photopolymerization-based bioprinting techniques, such as digital light processing (DLP) [7,14], have been used in manufacturing liver equivalents for organ-on-a-chip applications. DLP printing provided a high-resolution preparation of hydrogels with detailed architectures [7], and the mechanical properties of DLP products were improved by using poly(ethylene glycol) diacrylate (PEGDA) [15]. However, DLP printing requires potentially harmful photoinitiators in bioinks, and UV light associated with the photoinitiator affects cell behavior unexpectedly. Thus, a more promising liver-derived bioink needs to be developed to achieve good printability, cell viability, and functionality. Also, Chen et al. [14] developed a dECM-based bioink for building a thin hexagonal construct to serve as a platform for a hepatocellular carcinoma progression study using the DLP approach, but the construct printed was very thin (about 200 μm in height) and simple (hexagonal). Thus, how to fabricate a 3D thick bulk construct with high cell viability and functionality using DLP bioprinting with a dECM-based bioink is still a mystery. On the other hand, the applications of hepatocyte and HepG2 cells in liver tissue engineering are not convincing; HepG2, a tumor-derived cell, does not fully reflect the normal functioning of liver cells, and the limited sources of primary hepatocytes are far from meeting clinical applications. It is worth noting that a human-induced hepatocyte (hiHep cells) has been developed by Hui et al. [16,17], which is converted from fibroblasts by lentiviral expression of FOXA3, HNF1A, and HNF4A. Because the hiHep cell serves a biological function similar to the liver and can grow rapidly, it has become a promising candidate to replace the traditional liver cells in live tissue engineering, but 3D printing of hiHep cells has not been reported yet.

In this work, a microtissue was designed with an inner gear-like structure to achieve better cell function with a larger body surface area, and the microtissue was manufactured using DLP-based 3D bioprinting with cell-laden bioinks. Here, dECM was obtained from the decellularization of porcine liver tissues and was added to methacrylated gelatin (GelMA), forming GelMA/dECM cell-laden bioinks, which showed excellent mechanical properties, cytocompatibility, and printability. Then, hiHep cells were deposited with GelMA/dECM bioinks layer-by-layer, forming the liver microtissues. In the printed liver microtissues, dECM was found to improve hiHep cell viability and spreading. Moreover, hiHep cells maintained a stable living state with persistent liver function metabolism due to the addition of dECM in bioinks, and serve as a potential treatment for chronic liver failure or partial hepatectomy patients. This study not only provides a design and

manufacturing method for liver microtissues as a future liver donor to repair or even replace impaired liver but also develops a dECM-based bioink that is feasible for DLP printing with higher resolution and cell viability.

2. Materials and methods

2.1. Decellularization of porcine liver and dECM characterization

Fresh porcine liver tissues were harvested from Guangxi Bama mini pigs (*Sus scrofa*). Liver tissues were cut into small pieces and continuously rinsed in ultrapure water for 1 h. Then, samples were digested with 0.025% trypsin for 30 min, followed by treatment with 1% Triton solution for 24 h, and washed with ultrapure water for another 24 h. After soaking in 2% SDS for 36 h, samples of liver tissue became transparent. Finally, samples were washed with running water for 2–3 days to remove the detergents in the decellularized liver matrix and then were stored at 4 °C for further use. We used the Bama mini pig's (weighs about 25 kg) liver (about 350–500 g) for the decellularization process. Following the protocol, we eventually prepared a 50 mL 3% dECM solution.

Both native and decellularized pig liver tissues ($n = 3$) were fixed with 4% paraformaldehyde solution at room temperature for 24 h, followed by embedding the fixed tissues in paraffin. Paraffined sections were cut into thin slices of 5 μm for deparaffinization and rehydration. Then, cell component residues were observed by hematoxylin and eosin (H&E) staining. Collagen distributions were observed by Masson trichrome staining. Also, 5- μm -thick sections of native and decellularized liver tissues were stained using the nuclear stain, 4',6-diamidino-2-phenylindole (DAPI), in darkness under room temperature for 5 min. All the H&E staining and Masson trichrome staining samples were examined using an inverted microscope (Zeiss PrimoVert Inverted Microscope, Germany), and DAPI staining samples were examined using a fluorescent microscope (Olympus BX51, JAPAN).

DNA, glycosaminoglycan (GAG) and collagen content were measured using the kit respectively as the indicator of residual cells in the dECM following the manufacturer's instructions (DNA kit: CWBIO, CHINA; GAG kit: GENMED, CHINA; collagen kit: Chondrex, USA). A microplate reader (Thermo Scientific™ Multiskan™ GO, USA) was used for DNA, GAG and collagen content measurement.

2.2. Preparation of dECM bioink and DLP printing system

Lyophilized dECM was ground into powder, followed by digestion in 0.5 M of acetic acid and pepsin solution (10 mg pepsin per 100 mg dECM) for 4 days. After complete digestion, a dECM solution was obtained by centrifuging at 3000 rpm for 15 min to remove the particle. Then, 10 M of cold NaOH solution was added to the dECM solution to adjust the pH to approximately 7.4. The pH-adjusted dECM solution was stored at 4 °C for further use.

In this study, GelMA was synthesized as described in the previous literature [18], and lithium phenyl-2,4,6-trimethylbenzoylphosphine oxide (LAP) was synthesized and used as the photoinitiator. The GelMA was dissolved in PBS or dECM solution to form GelMA/dECM bioinks with a concentration of 10% (w/v) GelMA and 0.5% (w/v) LAP.

Before bioprinting, hiHep cells were digested by 0.125% trypsin-EDTA and centrifuged for 1 min. The GelMA/dECM bioinks were added, respectively, to dissolve the hiHep cells at a final density of approximately $2.5\text{--}3.0 \times 10^6$ cells mL^{-1} in accordance with the analysis, forming cell-laden bioinks. All the bioinks were warmed to 37 °C in the dark before use.

As shown in Fig. 1, a bottom-up DLP 3D printing system was developed that mainly consisted of a UV light source (Omnicure 2000) with a 365-nm wave length and 2.25 mW/cm^2 light intensity, a digital micromirror device (DMD) based on a Texas Instruments DLP module with a resolution of 1920×1080 , and a z-axis mobile platform. The

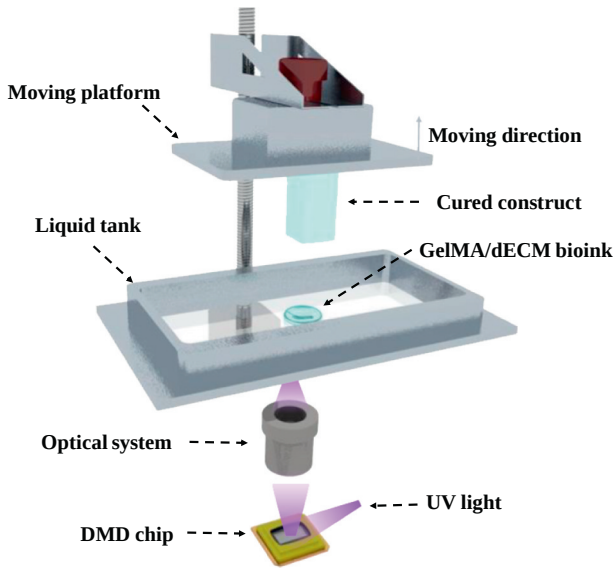


Fig. 1. The schematic of DLP-based 3D printer.

cell-laden GelMA/dECM bioinks were deposited in the material trough as a small droplet and cured by UV light reflected by the DMD chip.

2.3. Measurements of mechanical, swelling and degradation properties

The mechanical properties of hydrogels with different components were determined using the ElectroForce (TA Instruments, USA) testing machine at room temperature. Hydrogel samples with a rectangular shape of $20 \times 5 \times 1 \text{ mm}^3$ were tested using uniaxial stretch with a displacement rate of 1 mm/min, whereas hydrogel samples with a rectangular shape of $10 \times 5 \times 1 \text{ mm}^3$ were tested by unconfined compression at a displacement rate of 1 mm/min. Both Young's modulus and the compressive modulus of hydrogels were calculated by taking the slope of the initial linear part of the stress-strain curve [19].

The swelling property of GelMA hydrogels with and without dECM was measured by weighing the freeze-dried and swollen hydrogels. The hydrogel samples were manufactured in a cylindrical shape with a radius of 5 mm and a height of 0.5 mm by DLP printing and incubated in phosphate buffered saline (PBS) at 37 °C for 24 h. Then the samples were removed from the PBS, and the wet weight of the samples (W_s) was determined. The hydrogel samples were also lyophilized for 24 h and the dry weight (W_d) was determined. The swelling ratios of the hydrogels were obtained as

$$Q_s = \frac{W_s - W_d}{W_d} \quad (1)$$

The morphology of hydrogels and printed structures was observed using a scanning electron microscope (SEM, TM-1000, HITACHI). The samples were lyophilized using the lyophilizer (DYYM-10, Shanghai) for 24 h, and then sputter coated with iridium before scanning. The software ImageJ (version 1.8.0, National Institutes of Health, USA) was used to analyze the porosity based on SEM images.

The GelMA/dECM cylinders with 1 cm diameter and 3 mm height were obtained by curing bioinks with an ultraviolet lamp (365 nm) for 30 s, ensuring the same volume. All samples were lyophilized for 24 h and weighed to get the initial weight (W_0), and then soaked in the ethyl alcohol (75%) for 30 min. The sterilized samples were swelled in PBS at 37 °C for 24 h. The swelling samples were placed in a 24-well plate with enzyme solution (200 U/mL) and kept in a constant temperature incubator for degradation experiments, take samples ($n = 3$) every 6 h. Then, we lyophilize each sample for 24 h to weigh to obtain the remaining weight (W_r). The biodegradation property (Q_d) of hydrogels was indicated by the percentage of the remaining weight of the disks

after degradation as:

$$Q_d = \frac{W_r}{W_0} \times 100\% \quad (2)$$

2.4. Cell culture and analysis

The hiHep cells were cultured in hepatocyte maintenance medium (HMM) containing 500 mL DMEM/F12 (Gibco), 25 mL HMM mother liquor, 5 mL ITS (100×) (Gibco), 5 mL NeuroCult™ SM1 Neuronal Supplement (STEMCELL 05711), 0.5 mL TGF-α (40 μg/mL) (Perprotech af-100-16a), 0.5 mL EGF (40 μg/mL) (Perprotech af-100-15), and 50 uL Dexamethasone (10 mM) (Sigma D4902) as in the previous study [20]. The culture media was exchanged every two days.

The viability of hiHep cells encapsulated in microtissues was evaluated based on fluorescent observation using a LIVE/DEAD™ Viability/Cytotoxicity Kit (ThermoFisher L3224), while the migration of hiHep cells was evaluated using scratch test and phalloidin staining [21]. Both viability and phalloidin staining of hiHep cells were observed by a confocal microscope (Nikon A1 Ti, Japan). Cell proliferation after printing was measured with CCK8 after 1, 3, 5, and 7 days of culture, and the optical density (OD) values were determined using a microplate reader at 450 nm. The secretion of albumin and urea of hiHep cells in culture medium were determined using ELISA kit (Jymbio, Colorful-Gene Biological Technology Co., Ltd., Wuhan, China).

3. Results

3.1. Characterization of liver dECM

Upon decellularization, the liver tissues became translucent with the entire dissolution of the cells, and liver dECM powder was obtained after lyophilization (Fig. 2A). To assess the remaining cellular components in dECM, quantitative analysis of major biological components, histological analysis, and electron microscopy were performed.

DNA quantity in the decellularized liver matrix was observed to be significantly lower than in the native liver (Fig. 2B). Because the DNA contents exist in the nucleus of the cells, liver tissues were sufficiently decellularized. The glycosaminoglycan (GAG) (Fig. 2C) and collagen content (Fig. 2D) of the native and decellularized liver indicate a good biological property of dECM. The scanning electron microscopy (SEM) images of liver dECM ultrastructure are shown in Fig. 2(F), which reflect the porous characteristics of dECM. Furthermore, H&E and Masson staining show the absence of the cellular components in the decellularized liver matrix, while extracellular matrix components, such as collagen, are preserved (Fig. 2E). In addition, the DAPI staining of the decellularized liver matrix shows a very slight presence of nuclei comparing to the native liver tissue (Fig. 2E).

3.2. Characterization of GelMA/dECM bioinks

The influence of dECM on the mechanical properties of GelMA hydrogel is shown in Fig. 3(A–C). Based on the strain-stress curves of tensile (Fig. 3A) and compression tests (Fig. 3B), Young's modulus and compressive modulus are estimated as shown in Fig. 3C. It can be found that the compressive modulus and Young's modulus of the pure GelMA hydrogels and the GelMA/dECM hydrogel are similar. Furthermore, when comparing this information to the two moduli of native liver tissues, the hydrogels have a larger compressive modulus but a smaller Young's modulus than native liver tissues. It should also be noted that the native liver tissues experience a more ductile failure than GelMA/dECM hydrogels during the uniaxial stretch test.

The swelling ratio of GelMA/dECM hydrogels is shown in Fig. 3(D). The swelling ratio of GelMA is significantly increased by adding liver dECM. The swelling ratio of GelMA with dECM hydrogel is about 932%, which is much larger than the Swelling ratio of GelMA hydrogel (about

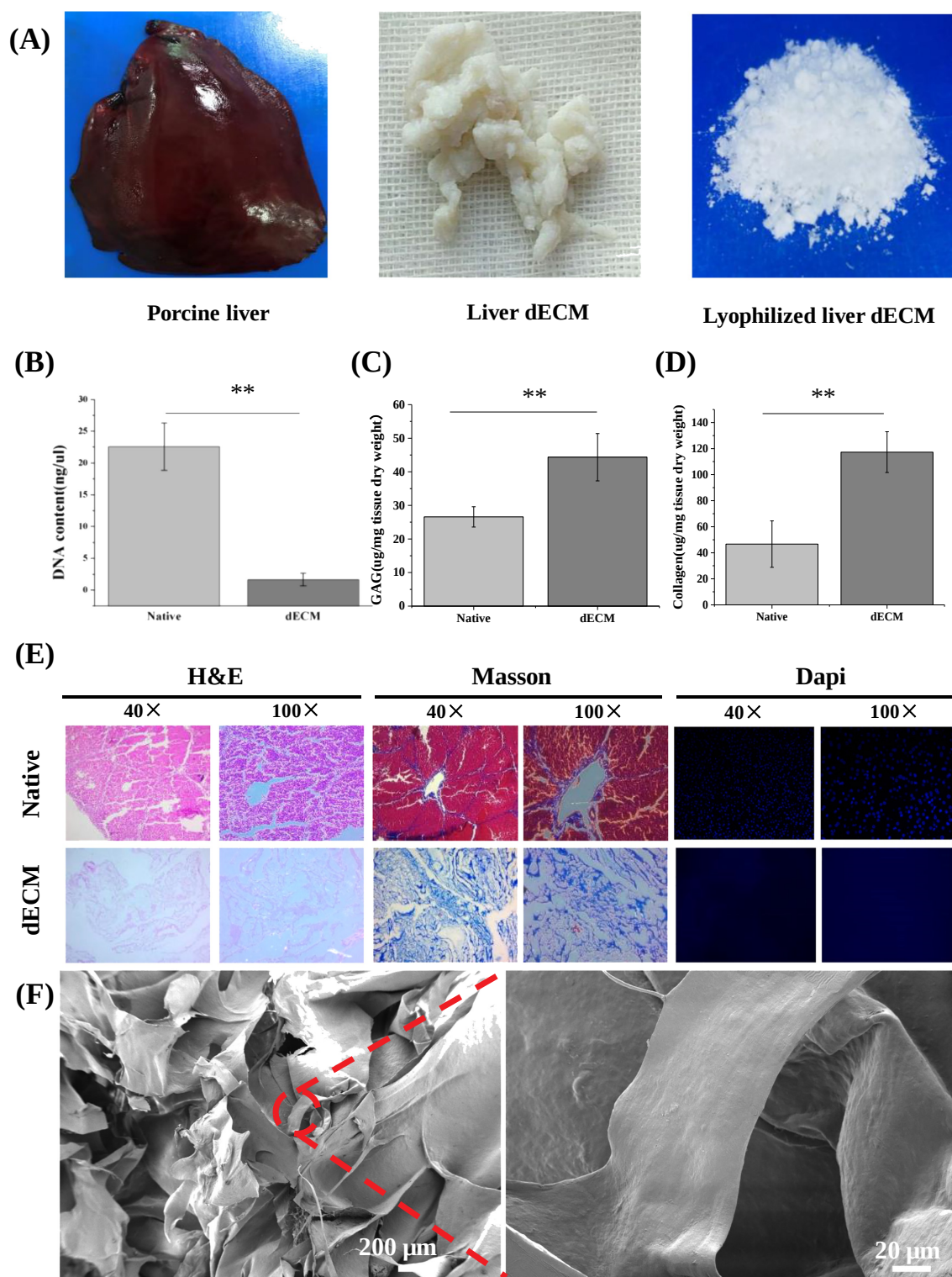


Fig. 2. Determination of liver decellularization. (A) Digital image of fresh porcine liver/liver dECM/lyophilized liver dECM. DNA quantification (B), GAG (C) and collagen content (D) of the native and decellularized liver. (E) H&E staining/Masson trichrome staining/nuclei staining of the native and decellularized liver tissues. Error bars represent standard error, p -values for native liver tissue versus dECM comparisons: * $p \leq 0.05$, ** $p \leq 0.01$. (F) The scanning electron microscopy of liver dECM ultrastructure.

644%).

The degradation curve of pure GelMA and GelMA with dECM in 24 h is shown in Fig. 3(E). It can be seen that GelMA hydrogel with dECM degrades more rapidly than pure GelMA hydrogel. This indicates that more protein and organic components are contained in dECM.

The cytocompatibility of GelMA/dECM bioinks was evaluated using

a culture of hiHep cells in bioinks. As shown in Fig. 4(A), all the bioinks have very good cell viability, more living cells are found in GelMA with dECM bioinks (both adding 1.5% or 3% dECM) after 1 and 3 days of incubation. Based on the CCK8 results in Fig. 4(B), the cell viability is promoted when dECM concentration is increased in GelMA with dECM bioinks. In addition, a scratch test of hiHep cells was carried out to

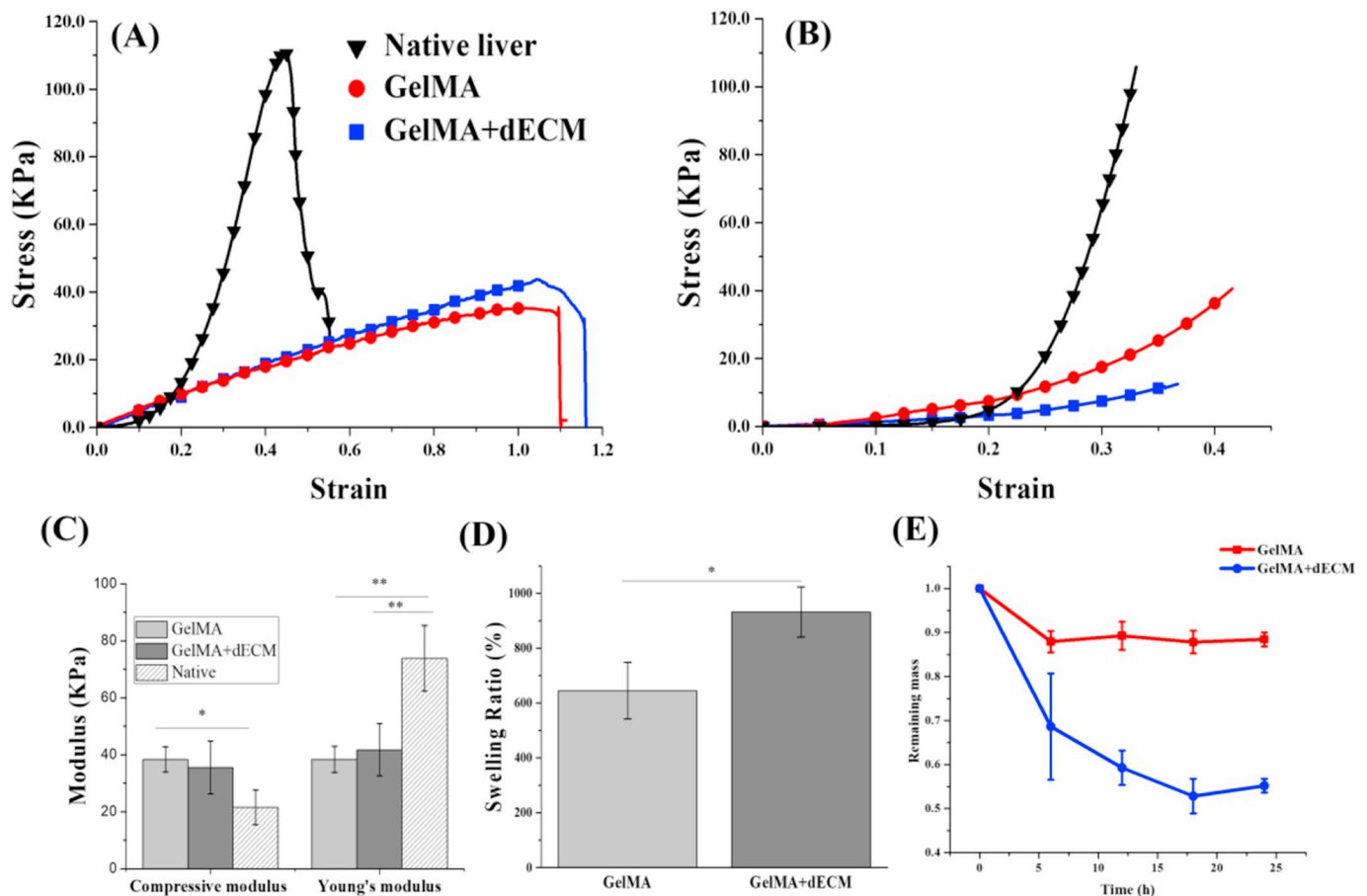


Fig. 3. Physical properties of native liver tissue, GelMA and GelMA/dECM hydrogels. The representative strain-stress curves of tensile (A) and compression (B) tests. (C) The compressive modulus and the Young's modulus. (D) Swelling ratio. * $p \leq 0.05$, ** $p \leq 0.01$. (E) The degradation curve of pure GelMA and GelMA with dECM under the action of enzyme.

detect the cell mobility on the surface of GelMA/dECM hydrogels. It was found that the addition of liver dECM in GelMA hydrogel significantly improved cell mobility on the hydrogel surface (Fig. 4C).

3.3. 3D bioprinting of liver microtissue using GelMA/dECM bioinks

Fig. 5(A) shows a CAD model of a radical microtissue with an inner gear shape, which is designed to increase the contact area of internal cells with the external environment, which improves hepatocyte function. The liver microtissue with a height of 2 mm was printed by DLP-based bioprinting using the hiHep cell-laden bioinks. Based on the curing test (Fig. S1 in SI), the curing time of 10% GelMA with 0.5% LAP is 3.5 s, and the curing time of GelMA with dECM is 2.5 s. Based on the microscopic observation of the inner structure of the liver microtissue, it can be found that the gear shape printed using GelMA with dECM bioink has more accurate rectangular angles than that printed using GelMA bioink (Fig. 5A), which indicates that the addition of liver dECM also improves the printing fidelity of GelMA/dECM bioinks. Also, a more regular gear shape structure was observed in the SEM image of GelMA with dECM microtissue (Fig. 5B). The SEM images show high porous morphology of liver microtissues, and the porosity of microtissues is estimated by software ImageJ (NIH, USA) based on SEM images. As shown in Fig. 5(C), the porosity of printed liver microtissues is significantly increased when GelMA with dECM bioink is used for printing.

The hiHep cells were embedded in the liver microtissues by printing of cell-laden bioinks, and the cylindrical cavity part of the microtissue was designed for circulation of culture medium, which is beneficial for

hiHep cell metabolism. The representative fluorescent images of living/dead cell staining within the printed microtissues are shown in Fig. 6(A), and the higher growth rate of hiHep cells is observed in the liver microtissues that were fabricated by GelMA/dECM bioinks after 1, 3, 5, and 7 days of cell culture. In addition, the dECM in GelMA bioinks is found to improve the cell viability in liver microtissues within 5 days of cell culture (Fig. 6D).

Cell spreading in the liver microtissues was visualized using SEM and found that the cells are more spreadable in microtissues with dECM (Fig. 6B). We also performed F-actin and DAPI staining under a confocal fluorescence microscope, which showed that more actin staining was found in hiHep cells of microtissues that were fabricated using GelMA/dECM hydrogels (Fig. 6C), which indicates more spreading morphology of hiHep cells in microtissues with liver dECM. Measuring the spreading area of hiHep cells in liver microtissues, it was also found that the size of hiHep cells in microtissues fabricated by GelMA/dECM bioinks is significantly larger than in GelMA bioinks (Fig. 6E), indicating better cell spreading in GelMA/dECM bioinks.

3.4. Liver-specific functional expression of microtissues

Although Fig. 6(A) demonstrates a higher living hiHep cell density in microtissues with liver dECM, the liver-specific cell functionality of hiHep cells in microtissues is more crucial for hepatic function recovery. Here, the ability of albumin and urea synthesis is chosen as two typical indices of hiHep cell functionality [1]. The secretion level of both albumin (ALB) and blood urea nitrogen (BUN) in culture supernatants is found to be significantly higher in microtissues with dECM

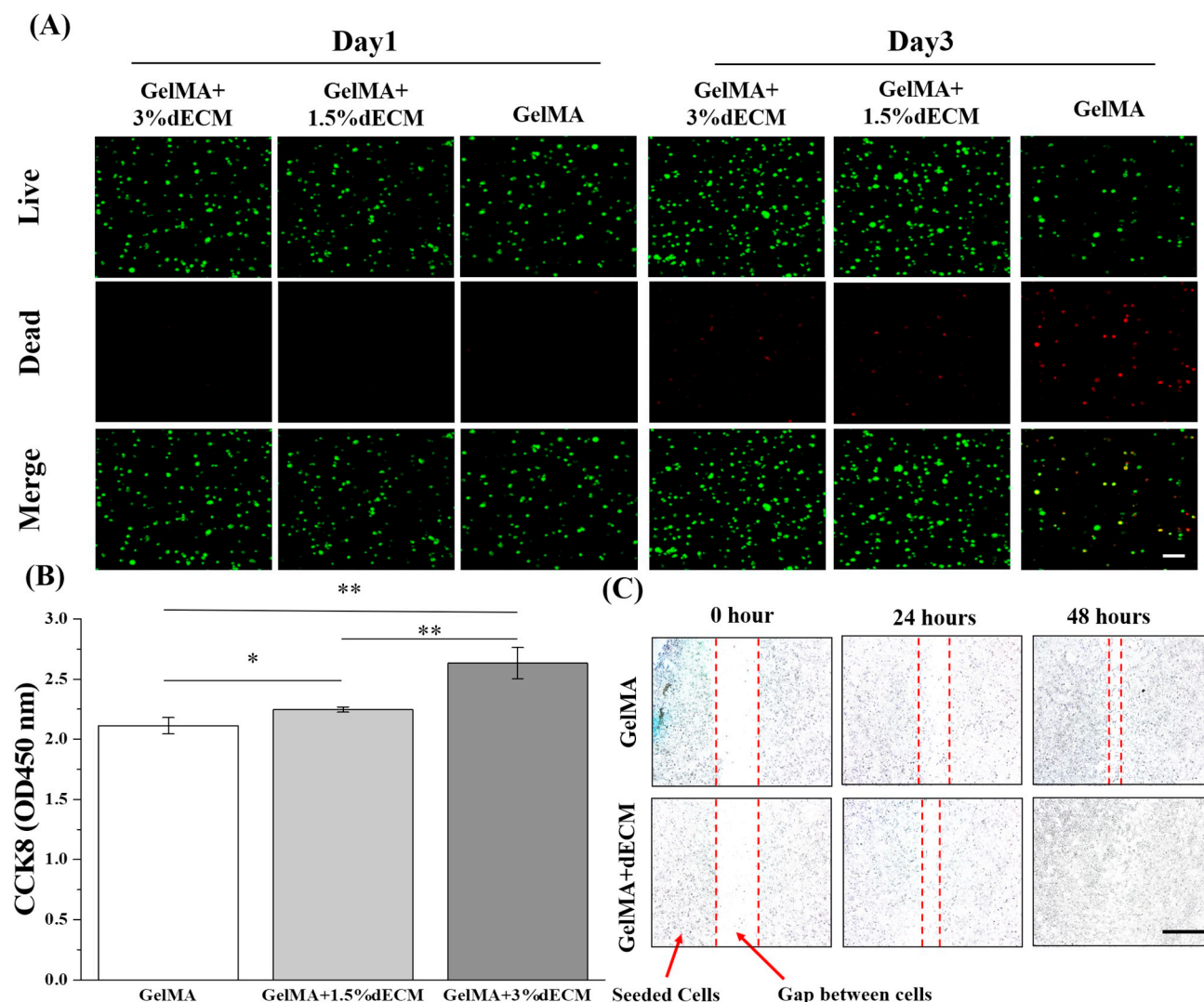


Fig. 4. Biological properties of native liver tissue, GelMA and GelMA/dECM hydrogels. (A) Representative live/dead fluorescence images of hiHep cells in GelMA/dECM bioinks after 1-day and 3-day of culture (scale bar: 100 μ m). (B) Cell proliferation of hiHep cells in GelMA/dECM hydrogels determined via a CCK-8 quantitative analysis after 1-day and 3-day of culture. (C) Cell migration of hiHep cells on GelMA/dECM hydrogel surface (scale bar: 500 μ m). * $p \leq 0.05$, ** $p \leq 0.01$.

than GelMA microtissues within 7 days of culture (Fig. 7). Also, it is noted that hiHep cells are unable to maintain a steady growth rate over 7 days after printing, because ALB secretion decreases after 5 days of culture.

4. Discussion

In recent studies, naturally derived dECM material-based bioinks have been developed for 3D bioprinting, and the majority of liver-specific dECM bioinks were used in extrusion-based 3D printing [6,8,15]; only very few studies [14,22] were reported to use liver-specific dECM bioinks in DLP-based 3D printing. In this study, it was found that adding liver dECM to GelMA bioink improves both printability and cell viability of GelMA-based bioinks, which would be of great importance for the application of dECM bioinks. Here, proteomic analysis and gene ontology analysis were carried out on liver dECM to understand its role in bioinks. As shown in Fig. 8(A), the proteinic compositions of ECM are classified according to their location and subdivided based on their abundance. It was found that the extracellular matrix and cytoskeleton are the major components of liver

dECM, but with very little nuclear or intracellular protein content. The extracellular matrix components, i.e., the collagen-containing extracellular matrix, exert biochemical and biomechanical functions on tissue morphogenesis, cell differentiation, and systemic homeostasis, which may contribute to the promotion of cellular growth and stimulation of cell activities of hiHep cells in liver dECM [23,24]. Also, we investigated the molecular function of liver dECM and identified the binding function as the pivotal role of liver dECM (Fig. 8B). With the proteome analysis of dECM, we detected an overwhelming quantity of functional proteins such as fibulin [25], integrins [26], and tenascin [27], which are capable of binding growth factors, interact with receptors to induce signal transduction and regulation of gene transcription, and assist in essential morphological organization and physiological functional regulation involved with cell growth, adhesion, and stretch. In addition, analysis of the biological process of dECM demonstrates that the dECM components mainly regulate multicellular organismal processes, cellular component organization, and biogenesis (Fig. 8C). Based on the above results, the liver dECM can provide a proper biological process by regulating the signal pathways for macromolecule biosynthesis, cellular component arrangement, and

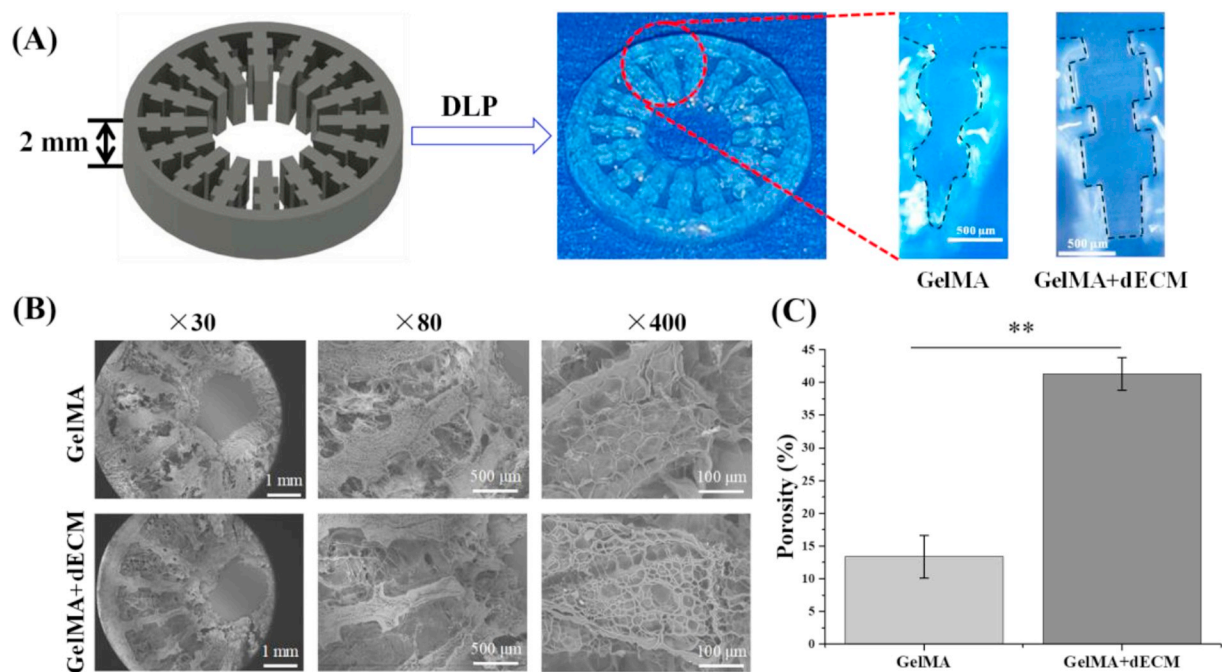


Fig. 5. The macroscopic and microscopic images of the DLP printed GelMA/dECM and GelMA scaffolds. (A) The designed liver microtissue model and the DLP printing results. (B) SEM images of printed liver microtissues. (C) The porosity of printed liver microtissues. * $p \leq 0.05$, ** $p \leq 0.01$.

disintegration involving genetic expression, protein modification, and intricate interaction with proteins and substrates, which thereby facilitates hiHep cell growth and differentiation [28,29]. Therefore, liver dECM in GelMA bioinks enhances growth (Fig. 6A), migration (Fig. 4C), spreading (Fig. 6E), and liver function maintenance (Fig. 7) of hiHep cells in printed liver microtissues.

Another interesting finding of the dECM effect is that the addition of liver dECM to GelMA bioinks significantly reduces the curing time of GelMA bioinks, and the curing time decreases with the dECM concentration in GelMA with dECM bioinks (Fig. S1 in SI). A shorter bioink curing time leads to better shape fidelity of microtissues (Fig. 5A) and provides higher viability of encapsulated cells in photocurable bioinks because of less UV light exposure time. Based on the results of mass spectrometry, which are consistent with previous research findings [30,31], collagen is found to be the main component of liver dECM. As reported by Berger et al. [32], adding collagen leads to the formation of an interpenetrating network in the GelMA hydrogels. Thus, liver dECM in GelMA bioinks would offer more photocrosslinking of sites to improve the crosslinking rate of bioinks [33], leading to better printability and cell viability of GelMA/dECM bioinks.

In previous studies of liver tissue engineering [34,35], it was preferred to seed human hepatocytes on the scaffolds, but this preference suffered from a shortage and poor liver function maintenance of human hepatocytes. Furthermore, human hepatoma cell lines, i.e., HepG2, were widely used in the studies of liver-specific dECM bioinks [14,36]. Although human hepatoma cell lines produce human proteins, these cells are tumor-derived cells and incompetent in metabolism and ammonia clearance [37]. Thus, in this study, hiHep cells, which were converted from human fibroblasts [20,38], were encapsulated in GelMA/dECM to form cell-laden bioinks. Although hiHep cells are considered as expandable cells, displaying better functional characteristics of mature hepatocytes compared to human hepatoma cell lines [38], few studies have been conducted on 3D cultures of hiHep cells, especially through 3D bioprinting. In our study, hiHep cells grew well

and exhibited a pre-determined cell spread in the 3D environment of printed liver microtissues, and liver dECM in the bioinks significantly enhanced the liver-specific functions of hiHep cells in printed liver microtissues. The development of liver dECM-based hiHep cell-laden bioinks may provide an effective way to establish the liver-specific function for tissue engineering products.

5. Conclusion

In this study, liver-specific bioinks for DLP-based printing were developed by combining liver dECM with GelMA; hiHep cells were then encapsulated to form cell-laden bioinks. Interestingly, liver dECM was found to not only improve the printability but also the hiHep cell viability of GelMA-based bioinks. Furthermore, a liver microtissue was manufactured by DLP using the liver-specific bioinks, and hiHep cells in liver microtissue expressed excellent hepatic functions. This work highlights the progress of liver-specific bioink development; the liver microtissue can be used as a liver substitute for patients with liver failure and provides an effective alternative to liver tissue engineering.

In future work, bioinks with multiple cells need to be developed, because the liver is a multicellular component organ, and the communication between different cells also plays an important role in liver tissue engineering. Moreover, an *in vivo* study needs to be carried out for liver microtissues with optimized geometry and functions to further help in understanding how to develop implantable liver substitutes and co-culture of multiple functional cells will also be performed.

CRedit authorship contribution statement

Qijiang Mao: Conceptualization, Data curation, Formal analysis, Investigation, Methodology. **Yifan Wang:** Data curation, Funding acquisition, Methodology, Project administration, Writing-original draft. **Yang Li:** Data curation, Formal analysis, Investigation. **Sarun Juengpanich:** Investigation, Writing-original draft. **Wenhuan Li:**

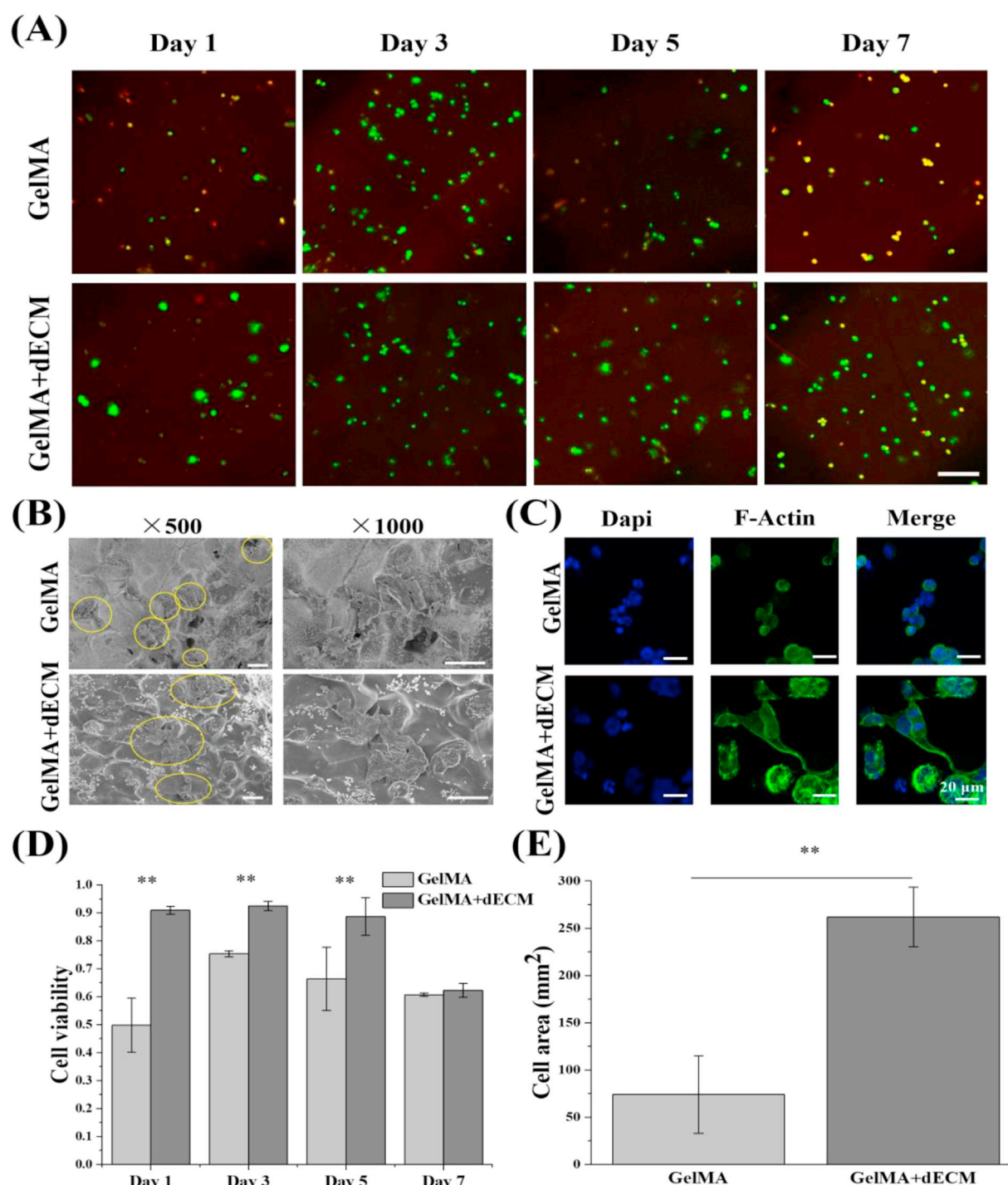


Fig. 6. The biological characteristics of hiHep cells seeded in GelMA and GelMA/dECM bioinks. (A) Representative live/dead fluorescence images of hiHep cells encapsulated in printed microtissues (scale bar: 100 μm). (B) SEM images of hiHep cells encapsulated in printed microtissues with 5 days culture (scale bar: 50 μm). (C) The cellular state of hiHep cells encapsulated in printed microtissues with 5 days culture shown by Phalloidin staining (scale bar: 20 μm). (D) Quantification of hiHep cell viability in printed microtissues. (E) Average cell area of hiHep cells encapsulated in printed microtissues with 5 days culture. * $p \leq 0.05$, ** $p \leq 0.01$.

Investigation. **Mingyu Chen:** Methodology. **Jun Yin:** Conceptualization, Funding acquisition, Writing-original draft. **Jianzhong Fu:** Conceptualization, Writing-review & editing. **Xiujun Cai:** Conceptualization, Funding acquisition, Writing-review & editing.

Declaration of competing interest

The authors whose names are listed immediately below certify that

they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

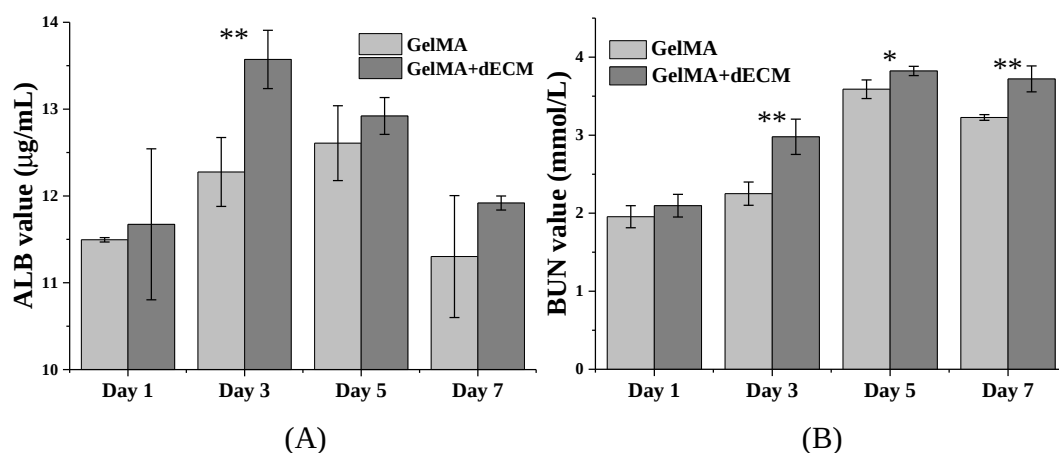


Fig. 7. Liver functional evaluation of hiHep cells seeded in GelMA/dECM bioinks after DLP printing. Albumin secretion levels (A) and blood urea nitrogen secretion levels (B) of hiHep cells encapsulated in printed liver microtissues. $n = 3$ for all data points. * $p \leq 0.05$, ** $p \leq 0.01$.

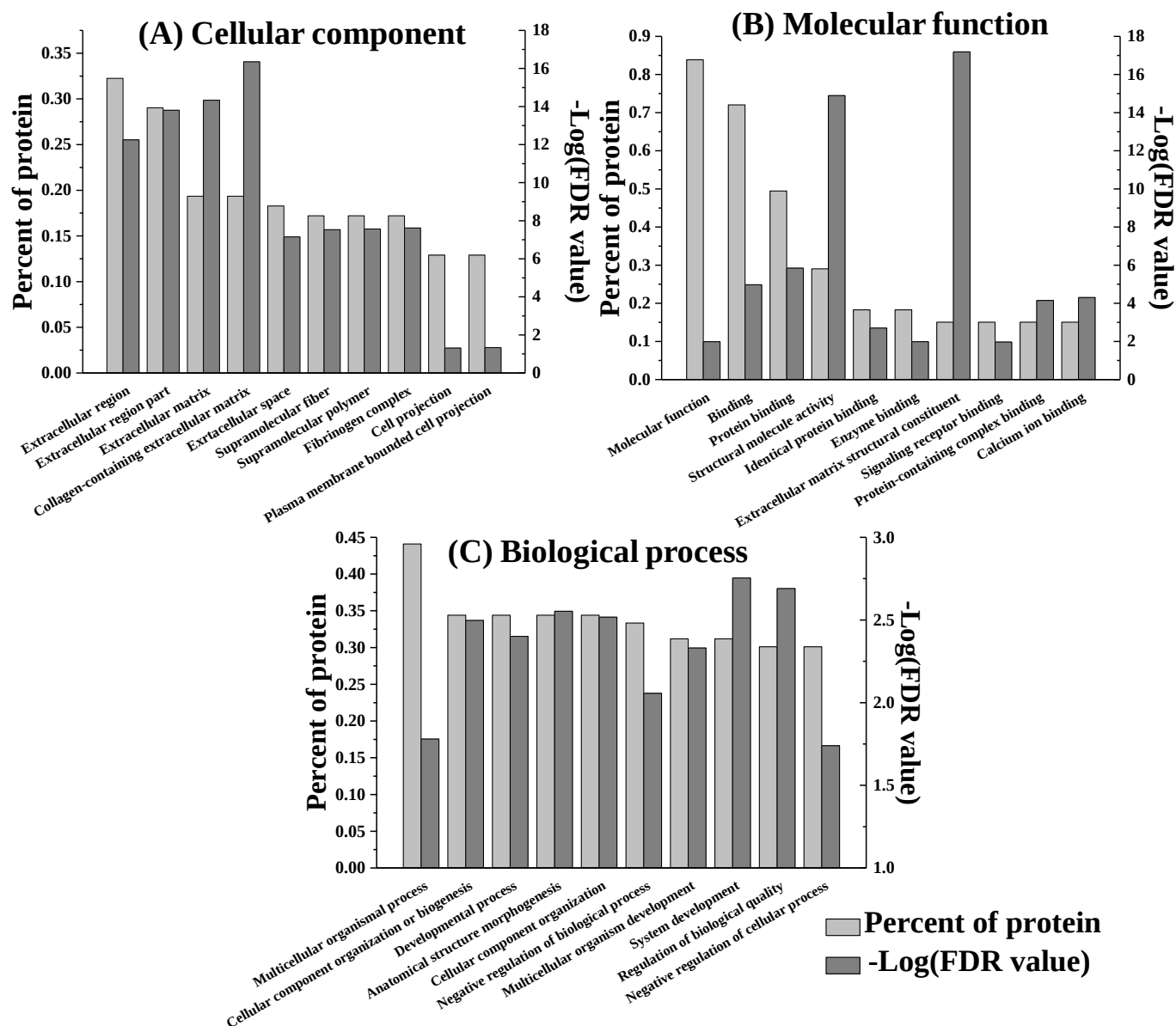


Fig. 8. Proteomic analysis of dECM. GO classification of dECM proteins in cellular component (A), molecular function (B), and biological process (C).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.msec.2020.110625>.

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